DATA ANALYTICS AND VISUALIZATION USING PYTHON

CASE STUDY REPORT

TOPIC

AIR QUALITY ANALYSIS, POLLUTION PREDICTION AND TREND FORECASTING USING TIME-SERIES FORECASTING (ARIMA)

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1. INTRODUCTION

Air Quality Analysis, Pollution Prediction and Trend Forecasting Using Indian Cities Data (2020–2024)

This project analyzes air-quality data collected from selected monitoring locations in major Indian cities from 2020 to 2024. The dataset contains measurements of PM2.5, PM10, NO_2, NH_3, SO_2, CO and O_3 along with date and location information.

The project uses Python, Excel, Power BI/Tableau and ARIMA to analyze pollution patterns, visualize pollutant trends and forecast future air-quality conditions.

2. PROBLEM STATEMENT

Air pollution varies over time and between different locations. Large amounts of pollutant data can make it difficult to identify:

- Which locations have higher pollution levels
- Which pollutants contribute significantly to poor air quality
- How pollution changes over time
- Seasonal or yearly pollution trends
- Future pollution levels

Therefore, this project aims to analyze historical air-quality data from 2020–2024 and use ARIMA time-series forecasting to predict future pollutant trends.

3. OBJECTIVES

Project objectives can be:

1. Analyze air-quality data from 2020–2024.
2. Compare pollution levels across 8 Indian monitoring locations.
3. Analyze trends of PM2.5, PM10, NO_2, NH_3, SO_2, CO and O_3.
4. Identify yearly and time-based pollution patterns.
5. Perform data cleaning and preprocessing.

6. Create meaningful visualizations using Python, Excel, Power BI and/or Tableau.
7. Apply ARIMA for time-series forecasting.
8. Forecast future pollutant concentrations.
9. Compare historical pollution trends with predicted values.
10. Present the findings using dashboards and visual reports.

4. DATASET DESCRIPTION

Dateset Name: Air_Quality_analysis.csv

Dataset has these columns:

Number of Rows:14616

Number of Columns : 9

Column	Description
Timestamp	Date of observation, e.g. 01-01-2020
Location	Air-quality monitoring location
PM2.5	Particulate matter ≤ 2.5 µm
PM10	Particulate matter ≤ 10 µm
NO2	Nitrogen dioxide
NH3	Ammonia
SO2	Sulfur dioxide
CO	Carbon monoxide
O3	Ozone

Locations

Dataset contains these 8 locations:

1. Chennai – Alandur Bus Depot
2. Bengaluru – Silk Board
3. Delhi – Punjabi Bagh
4. Hyderabad – Central University
5. Kolkata – Rabindra Bharati University
6. Lucknow – Lalbagh
7. Mumbai – Chhatrapati Shivaji International Airport (T2)
8. Visakhapatnam – GVM Corporation

Time period: 2020–2024

5. DATA PREPROCESSING

This section explains how I prepared the dataset before performing the analysis

Include:

- **Handle missing values (NA)**

  print(df.isnull().sum())

```
In [5]: print(df.isnull().sum())
Timestamp       0
Location        0
PM2.5         346
PM10          916
NO2           367
NH3          1036
SO2           424
CO            299
O3            407
dtype: int64
```

  Handle NA and empty cells in PM2.5, PM10, NO2, NH3, SO2, CO, O3

  # Replace NA and empty values with NaN

df = df.replace(["NA", "N/A", "na", "n/a", "", " "], np.nan)

df = df.replace(r'^\s*$', np.nan, regex=True)

Interpolate missing pollutant values based on each location for column in pollutants:

if column in df.columns:

df[column] = (df.groupby("Location")[column] .transform(lambda x: x.interpolate(method="linear", limit_direction="both")))

```
In [8]: print(df.isnull().sum())
Timestamp    0
Location     0
PM2.5        0
PM10         0
NO2          0
NH3          0
SO2          0
CO           0
O3           0
dtype: int64
```

- **Remove duplicate records**

```
In [9]: print("Duplicate rows:", df.duplicated().sum())
Duplicate rows: 0
```

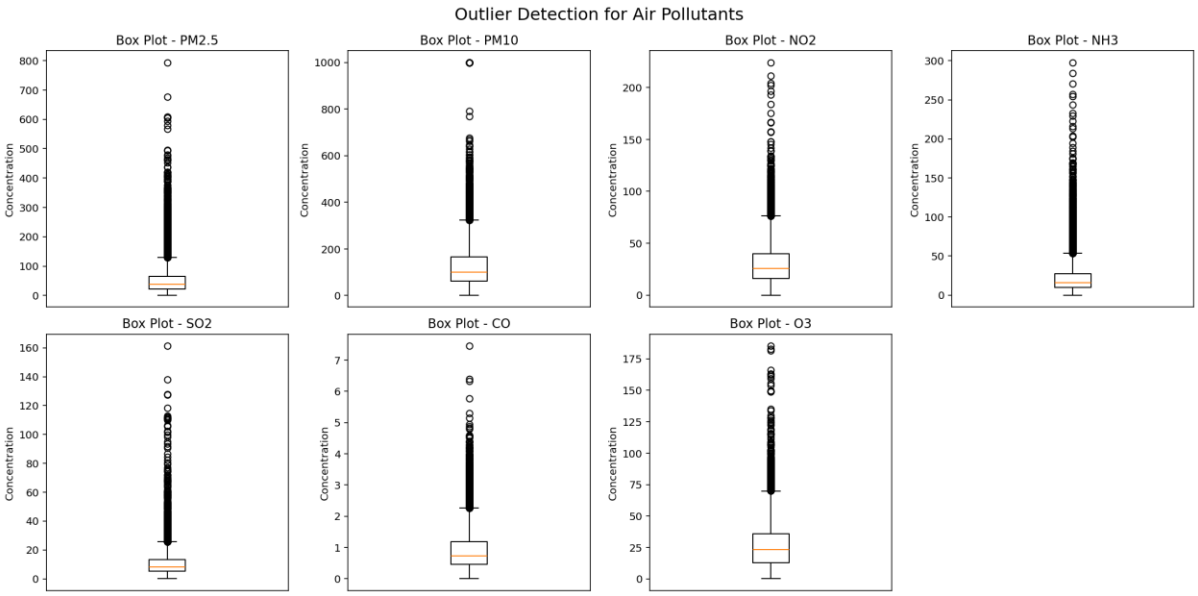
- **Correct inconsistent data**

```
In [14]: print(df["Location"].unique())
<ArrowStringArray>
[                    'Chennai - Alandur Bus Depot',
                         'Bengaluru - Silk Board',
                           'Delhi - Punjabi Bagh',
                  'Hyderabad - Central University',
            'Kolkata - Rabindra Bharati University',
                                'Lucknow - Lalbagh',
 'Mumbai - Chhatrapati Shivaji Intl. Airport (T2)',
                 'Visakhapatnam - GVM Corporation']
Length: 8, dtype: string
```

- **Detect and handle outliers**

```
import matplotlib.pyplot as plt
for column in pollutants:
    if column in df.columns:
        Q1 = df[column].quantile(0.25)
        Q3 = df[column].quantile(0.75)
        IQR = Q3 - Q1
        lower = Q1 - 1.5 * IQR
        upper = Q3 + 1.5 * IQR
        outliers = df[
            (df[column] < lower) |
            (df[column] > upper)
        ]
        print(column, "Outliers:", len(outliers))
        plt.figure(figsize=(6, 4))
        plt.boxplot(df[column].dropna())
        plt.title(f"Box Plot - {column}")
        plt.ylabel(column)
        plt.show()
```

```
PM2.5 Outliers: 1138
PM10 Outliers: 698
NO2 Outliers: 476
NH3 Outliers: 1084
SO2 Outliers: 737
CO Outliers: 621
O3 Outliers: 415
```



- **Convert data types where necessary**

  Timestamp → datetime

  df["Timestamp"] = pd.to_datetime(df["Timestamp"],format="%d-%m-%Y",

  errors="coerce")

  print(df.dtypes)

```
Timestamp    datetime64[us]
Location                str
PM2.5               float64
PM10                float64
NO2                 float64
NH3                 float64
SO2                 float64
CO                  float64
O3                  float64
dtype: object
```

df["Average_Pollution"] = df[pollutants].mean(axis=1)

print("\n--- AVERAGE POLLUTION FEATURE ---")

print(df[["Timestamp", "Location", "Average_Pollution"]].head())

```
--- AVERAGE POLLUTION FEATURE ---
   Timestamp                      Location  Average_Pollution
0 2020-01-01  Chennai - Alandur Bus Depot             14.082
1 2020-01-02    Chennai - Alandur Bus Depot           14.818
2 2020-01-03    Chennai - Alandur Bus Depot           18.832
3 2020-01-04    Chennai - Alandur Bus Depot           16.968
4 2020-01-05    Chennai - Alandur Bus Depot           20.638
```

6. EDA – EXPLORATORY DATA ANALYSIS

This is where you investigate the dataset before applying ARIMA.

Descriptive Statistics of Air Quality Data

	PM2.5	PM10	...	CO	O3
count	14270.000000	13700.000000	...	14317.000000	14209.000000
mean	54.768000	122.447124	...	0.911380	26.427832
std	54.452797	88.504204	...	0.647224	18.352948
min	1.280000	1.970000	...	0.000000	0.070000
25%	21.990000	60.270000	...	0.460000	12.800000
50%	38.375000	97.965000	...	0.730000	23.130000
75%	65.987500	157.540000	...	1.190000	35.590000
max	794.000000	999.990000	...	7.450000	185.190000

Formulate and answer at least 5 analytical questions.

Q1. Which location has the highest average concentration of each pollutant?

Compare average PM2.5, PM10, NO_2, etc. between the 8 locations.

It compares average pollutant levels across different locations. PM10 and PM2.5 are generally higher, with the highest PM10 level at Delhi – Punjabi Bagh. Overall, pollution levels vary by location and pollutant type.

Average Pollution by Location:

Location	PM2.5	PM10	NO2	NH3	SO2	CO	O3
Bengaluru - Silk Board	31.978198	85.249743	24.395776	12.836140	5.508054	0.709047	27.743187
Chennai - Alandur Bus Depot	38.437614	69.733963	22.976644	19.366557	10.913060	1.088523	20.443916
Chennai - Alandur Bus Depot	29.550000	NaN	17.410000	NaN	3.740000	0.520000	19.190000
Delhi - Punjabi Bagh	114.174567	212.470714	47.171755	23.287223	17.082618	1.223433	28.134715
Hyderabad - Central University	29.064287	75.612365	34.053559	13.357052	4.895498	0.451715	22.328027
Kolkata - Rabindra Bharati University	59.200404	119.620834	20.468420	23.334831	14.781307	0.697439	39.128648
Lucknow - Lalbagh	83.072111	172.682093	33.684686	61.374331	7.415432	1.496713	32.555558
Mumbai - Chhatrapati Shivaji Intl. Airport (T2)	34.492232	122.362238	25.011713	15.636881	11.922513	0.996838	19.666138
Visakhapatnam - GVM Corporation	45.471384	111.917526	35.008462	16.158320	11.465140	0.626237	21.787804

Q2. How do the average pollutant levels change from 2020 to 2024?

Compare:

2020 → 2021 → 2022 → 2023 → 2024

The table shows average pollution levels from 2020 to 2024. PM2.5 peaked in 2021, while PM10 and NO_2 peaked in 2023. Overall, pollution levels varied across the five years.

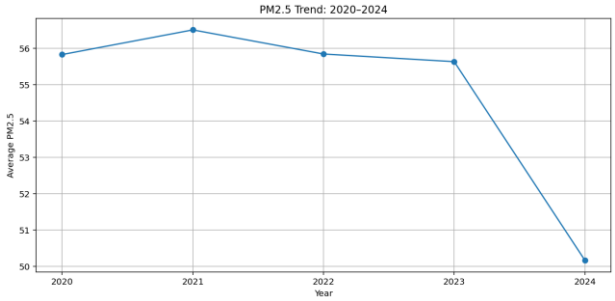
Average Pollution by Year:

Year	PM2.5	PM10	NO2	NH3	SO2	CO	O3
2020	55.823395	119.961080	29.803563	18.480475	8.810771	0.769120	30.264758
2021	56.501543	121.831790	32.239419	22.835460	12.125488	0.947305	31.771189
2022	55.838771	131.494229	31.520535	22.397539	11.579118	0.916628	26.724637
2023	55.627796	124.204659	34.679368	25.024108	9.848450	1.059937	23.665588
2024	50.163663	114.421045	24.036286	24.938885	10.156146	0.862660	19.770060

Q3. How does PM2.5 change from 2020 to 2024?

→ Use a PM2.5 yearly trend.

- PM2.5 changes



Q4. Which pollutants show increasing or decreasing trends between 2020 and 2024?

It show increasing or decreasing trends between 2020 and 2024?

```
PM10 -> Decreasing
NO2 -> Decreasing
NH3 -> Increasing
SO2 -> Increasing
CO -> Increasing
O3 -> Decreasing
```

Q5. What are the relationships among the major air pollutants?

Study relationships between pollutants.

For example:

PM2.5 ↔ PM10

NO_2 ↔ CO

PM2.5 ↔ NO_2

```
Correlation Matrix:
          PM2.5      PM10       NO2       NH3       SO2        CO        O3
PM2.5  1.000000  0.892140  0.498265  0.313457  0.191358  0.478885  0.148471
PM10   0.892140  1.000000  0.520903  0.344424  0.208710  0.490465  0.154579
NO2    0.498265  0.520903  1.000000  0.217914  0.031448  0.361116  0.074531
NH3    0.313457  0.344424  0.217914  1.000000 -0.000319  0.425250  0.095145
SO2    0.191358  0.208710  0.031448 -0.000319  1.000000  0.096230  0.080589
CO     0.478885  0.490465  0.361116  0.425250  0.096230  1.000000  0.070710
O3     0.148471  0.154579  0.074531  0.095145  0.080589  0.070710  1.000000
```

The correlation matrix shows that PM2.5 and PM10 have the strongest positive correlation (0.892), meaning they vary closely together. NO_2, CO, and NH_3 show moderate relationships with particulate pollutants, while O_3 and SO_2 have mostly weak correlations with the other pollutants. Overall, particulate matter shows the strongest associations.

Findings obtained from the analysis.

Finding 1

Location: The analysis shows that Delhi – Punjabi Bagh recorded the highest average PM2.5 concentration (114.17), while Hyderabad – Central University recorded the lowest average PM2.5 concentration (29.06) among the locations analyzed.

Finding 2

Year: The yearly analysis shows that the average PM2.5 concentration decreased overall from 55.82 in 2020 to 50.16 in 2024, although it increased slightly in 2021 before declining.

Finding 3

PM2.5 Trend: PM2.5 showed a fluctuating but overall decreasing trend during the study period, with the highest average value occurring in 2021 (56.50).

Finding 4

Other Pollutants: PM10, NO_2, NH_3, SO_2, CO and O_3 showed different patterns across the five-year period. NH_3, SO_2 and CO showed an increasing trend, while PM10, NO_2 and O_3 showed a decreasing trend from 2020 to 2024.

Finding 5

Relationships: The correlation analysis indicates that PM2.5 and PM10 have a strong positive relationship (r = 0.892), while CO and O_3 show a very weak positive relationship (r = 0.071).

7. MACHINE LEARNING

Time-Series Forecasting Using ARIMA

Time-series forecasting is used to predict future air-pollutant concentrations based on historical observations. In this project, the ARIMA (AutoRegressive Integrated Moving Average) model is used to forecast pollutant levels such as PM2.5.

The historical data from 2020–2024 is arranged chronologically and divided into training and testing data. The ARIMA model is then trained on the historical values and used to predict future concentrations. Model performance is evaluated using MAE, RMSE, and MAPE.

Process

Historical Data → Data Preprocessing → Time-Series Creation → Train/Test Split → ARIMA Model → Forecast → Evaluation

CODE:

```
for location in locations:

    print("Location:", location)
    # Select location
    location_data = df[
        df["Location"] == location
    ].copy()
    # Select required columns
    location_data = location_data[
```

```
    ["Timestamp", "PM2.5"]
].dropna()

# Sort by date
location_data = location_data.sort_values(
    "Timestamp"
)

# Handle duplicate dates
location_data = (
    location_data
    .groupby("Timestamp", as_index=False)["PM2.5"]
    .mean()
)

# Set Timestamp as index
location_data.set_index(
    "Timestamp",
    inplace=True
)

# Set daily frequency
location_data = location_data.asfreq("D")

# Fill missing PM2.5 values
location_data["PM2.5"] = (
    location_data["PM2.5"]
    .interpolate(method="linear")
)

# Check data
if len(location_data) < 100:
    print("Not enough data for:", location)
    continue


# TRAIN / TEST SPLIT


train_size = int(
    len(location_data) * 0.80
)

train = location_data["PM2.5"].iloc[:train_size]

test = location_data["PM2.5"].iloc[train_size:]



# BUILD ARIMA MODEL
```

```
    model = ARIMA(
        train,
        order=(1, 1, 1)
    )

    model_fit = model.fit()
    # FORECAST TEST DATA
    forecast = model_fit.forecast(
        steps=len(test)
    )

    # Make forecast index same as test
    forecast.index = test.index

    # EVALUATION
    mae = mean_absolute_error(
        test,
        forecast
    )

    rmse = np.sqrt(
        mean_squared_error(
            test,
            forecast
        )
    )
    # Avoid division by zero
    non_zero = test != 0
    mape = (
        np.mean(
            np.abs(
                (test[non_zero] - forecast[non_zero])
                / test[non_zero]
            )
        ) * 100
    )
    print("Training data :", len(train))
    print("Testing data  :", len(test))
    print("MAE           :", round(mae, 3))
    print("RMSE          :", round(rmse, 3))
    print("MAPE          :", round(mape, 3), "%")

# 11. FINAL RESULTS TABLE
results_df = pd.DataFrame(results)
print("ARIMA RESULTS - ALL LOCATIONS")
print(
    results_df.to_string(index=False)
)
```

OUTPUT:

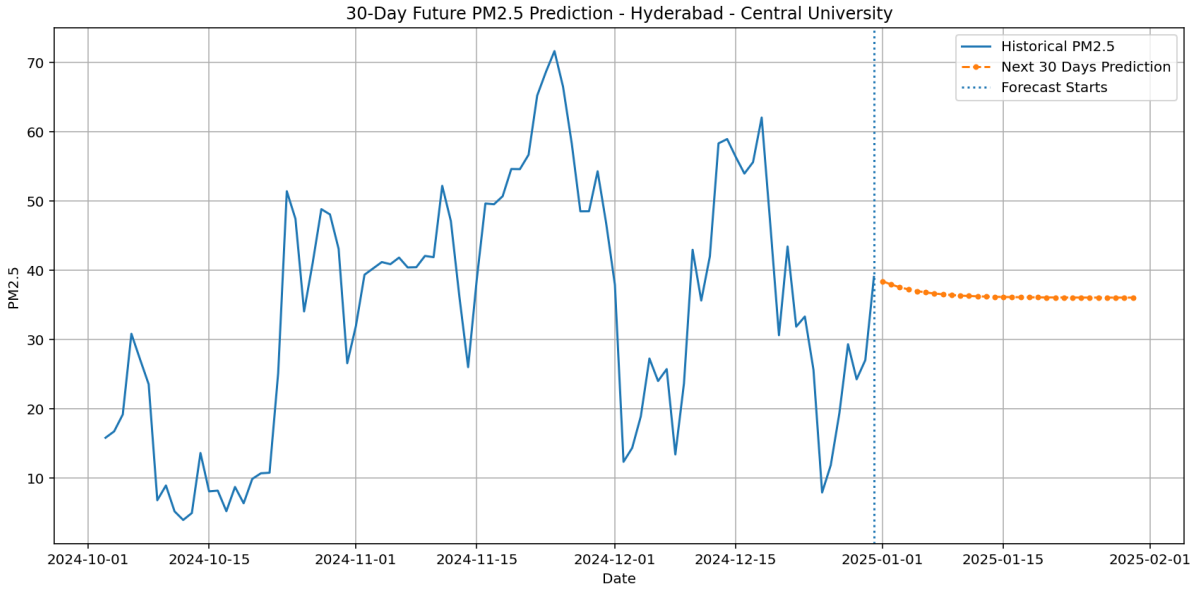
```
Location: Chennai - Alandur Bus Depot
Training data : 1461
Testing data  : 366
MAE           : 27.603
RMSE          : 35.542
MAPE          : 96.054 %
Location: Bengaluru - Silk Board
Training data : 1460
Testing data  : 366
MAE           : 21.643
RMSE          : 47.558
MAPE          : 117.277 %
Location: Delhi - Punjabi Bagh
Training data : 1461
Testing data  : 366
MAE           : 167.036
RMSE          : 181.224
MAPE          : 273.975 %
Location: Hyderabad - Central University
Training data : 1461
Testing data  : 366
MAE           : 19.179
RMSE          : 21.492
MAPE          : 181.584 %
```

```
ARIMA RESULTS - ALL LOCATIONS
                           Location        MAE        RMSE       MAPE
         Chennai - Alandur Bus Depot  27.602574  35.542394  96.054119
              Bengaluru - Silk Board  21.643338  47.558337 117.277184
                Delhi - Punjabi Bagh 167.036267 181.224170 273.974730
       Hyderabad - Central University  19.178833  21.492241 181.584114
Kolkata - Rabindra Bharati University  34.827364  38.971384 221.161308
                    Lucknow - Lalbagh  52.796000  59.668344 149.245543
      Visakhapatnam - GVM Corporation  40.380079  44.252149 141.944530
```

- ARIMA(1,1,1) was applied to 8 locations, with valid results for 7 locations.
- Mumbai was excluded due to insufficient data.
- Hyderabad and Bengaluru showed lower prediction errors.
- Delhi showed the highest prediction error.
- Overall, the high MAPE values suggest that the ARIMA(1,1,1) model needs improvement for accurate PM2.5 forecasting.

After evaluation, a new ARIMA model was trained using the complete historical data to predict PM2.5 levels for the next 30 days. The predicted values were saved in CSV files along with their corresponding dates and locations.

Graphs were generated to compare recent historical PM2.5 measurements with the predicted values. The evaluation results and future predictions were also combined into separate CSV files for further analysis.

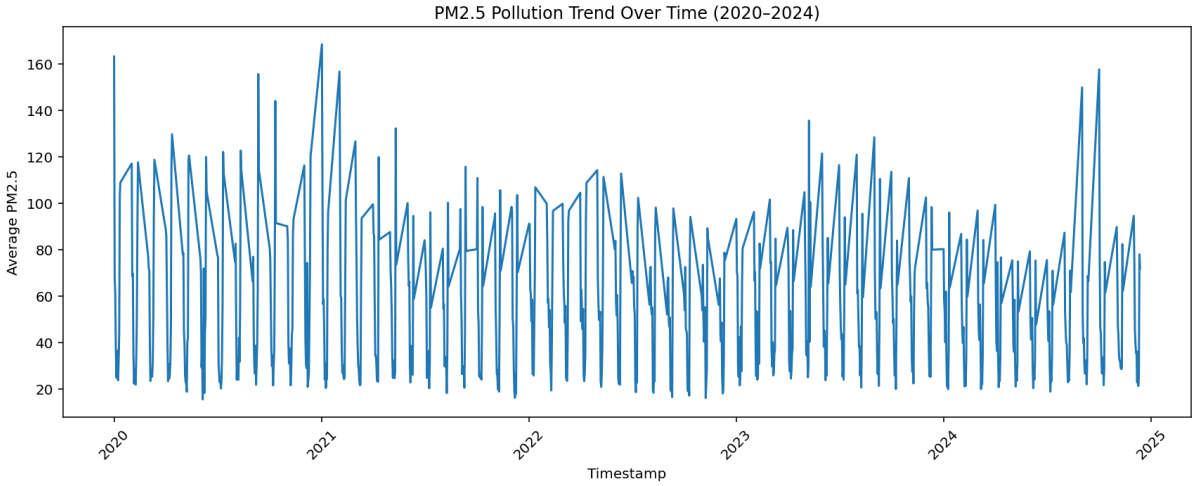


8. PYTHON VISUALIZATIONS

Visualization 1 – Pollution Trend Over Time

Line chart

Purpose: Identify how pollution changes over time.

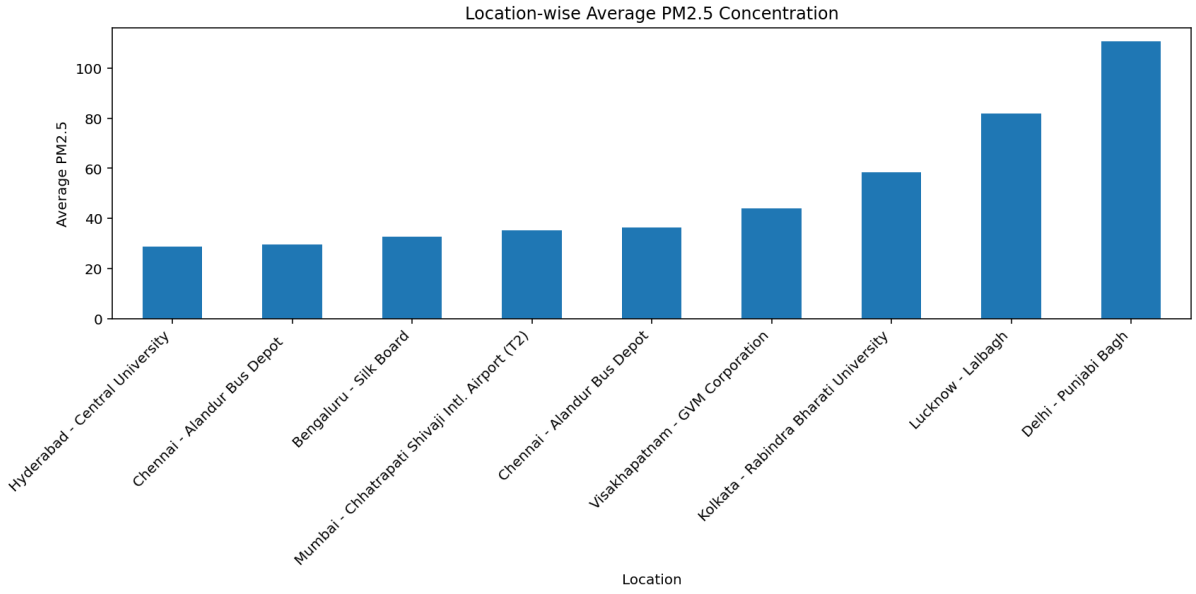


Visualization 2 – Location-wise Pollution Comparison

Bar chart

Compare all 8 locations.

Purpose: Identify differences in pollution levels between monitoring locations.

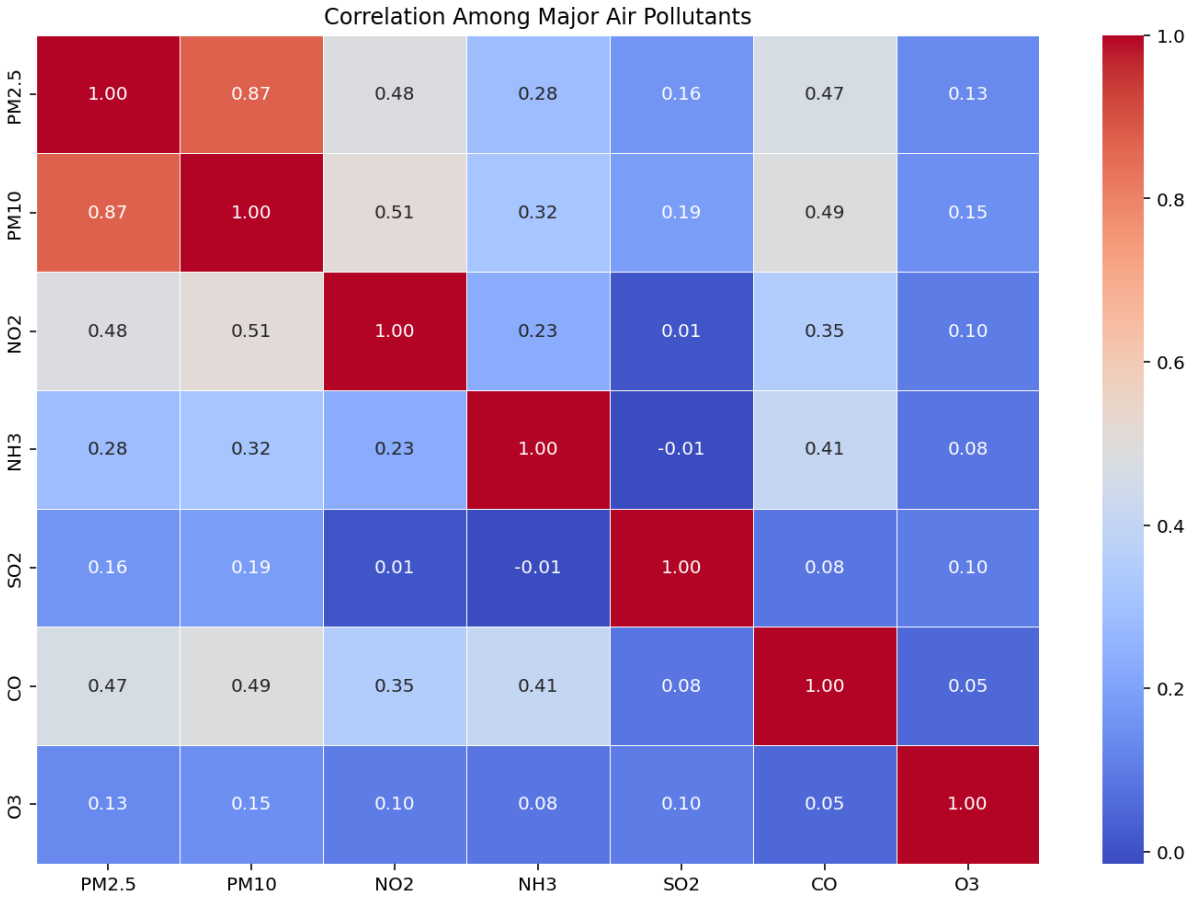


Visualization 3 – Pollutant Correlation Heatmap

Use:

PM2.5, PM10, NO_2, NH_3, SO_2, CO, O_3

Purpose: Identify relationships between different pollutants



Visualization 4 – Year-wise Pollution Comparison

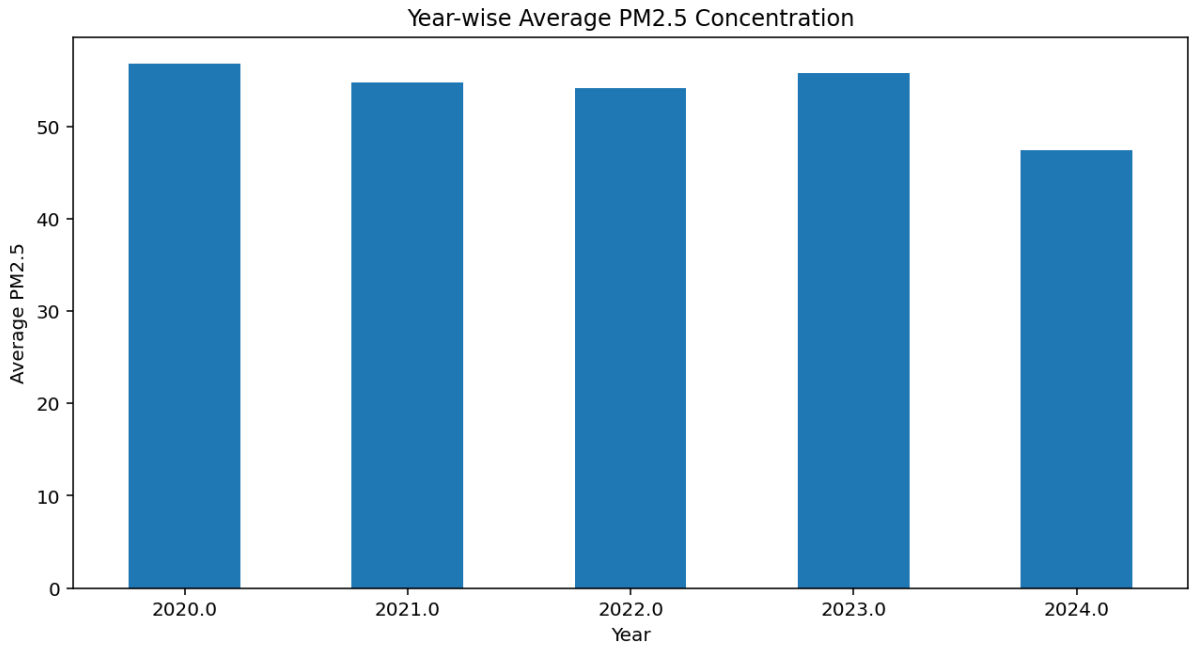
bar chart

Compare:

2020 | 2021 | 2022 | 2023 | 2024

for PM2.5 or another selected pollutant.

Purpose: Understand yearly variation and distribution.



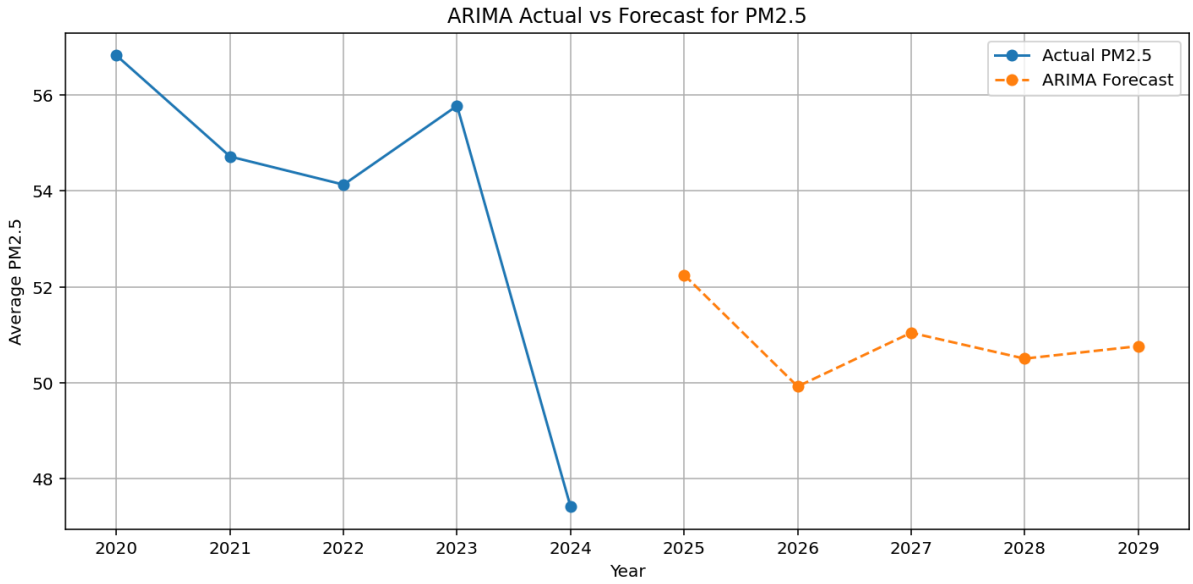
Visualization 5 – ARIMA Actual vs Forecast

Line chart

Show:

Actual PM2.5 and ARIMA Forecast

Purpose: Demonstrate the forecasting component of your project.



9. POWER BI DASHBOARD

The air-quality dataset contains pollution measurements from different Indian locations between 2020 and 2024. Microsoft Power BI was used to visualize pollution levels, yearly trends, and relationships between pollutants after cleaning and preparing the data.

Chart 1: Average PM2.5 by Location

Problem Statement: Which location has the highest average PM2.5 level?

Description: A clustered bar chart compares average PM2.5 concentrations across eight monitoring locations.

Purpose: To identify locations with higher and lower pollution levels.

Chart 2: Year-wise Average PM2.5 Trend

Problem Statement: How did PM2.5 levels change from 2020 to 2024?

Description: A line chart displays yearly average PM2.5 concentrations to show changes over time.

Purpose: To identify increasing or decreasing pollution trends.

Chart 3: Relationship Between PM2.5 and PM10

Problem Statement: What relationship exists between PM2.5 and PM10?

Description: A scatter chart compares PM2.5 and PM10 values to reveal patterns and relationships between the pollutants.

Purpose: To identify correlations and unusual observations.

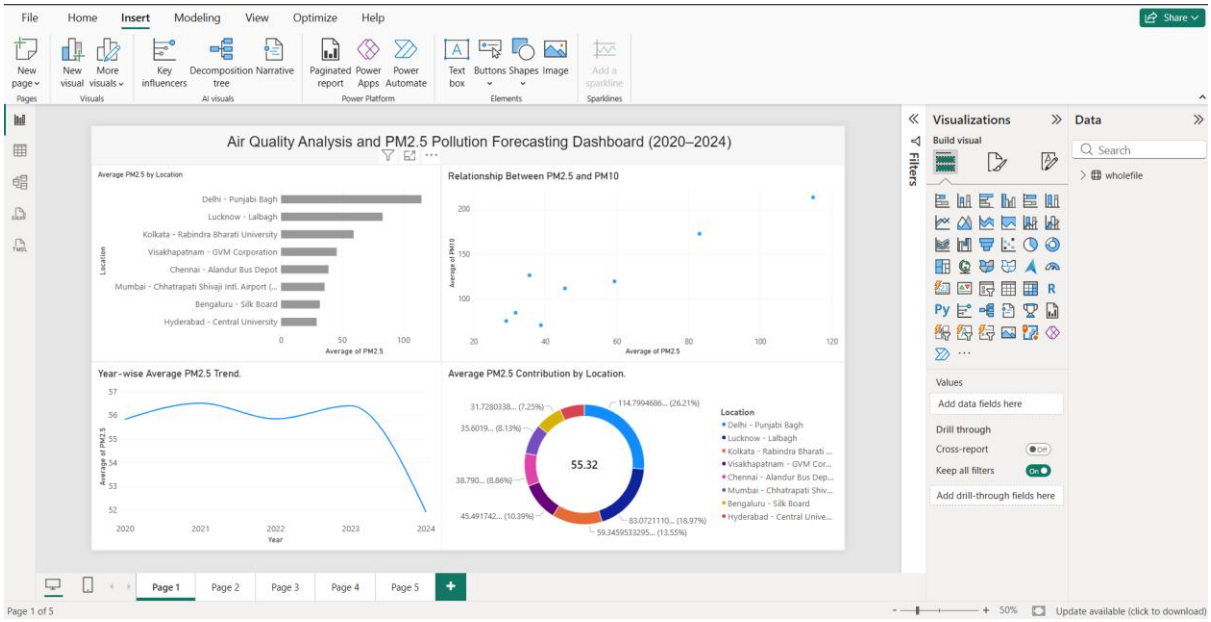
Chart 4: Average PM2.5 Using a Donut Chart

Problem Statement: How do average PM2.5 levels compare across locations?

Description: A donut chart represents each location's average PM2.5 level using different coloured segments.

Purpose: To provide a quick visual comparison of pollution levels.

OUTPUT:


Air Quality Analysis and PM2.5 Pollution Forecasting Dashboard (2020–2024)
Average PM2.5 by Location
Delhi - Punjabi Bagh
Lucknow - Lalbagh
Kolkata - Rabindra Bharati University
Visakhapatnam - GVM Corporation
Chennai - Alandur Bus Depot
Mumbai - Chhatrapati Shivaji Intl. Airport (...
Bengaluru - Silk Board
Hyderabad - Central University
Relationship Between PM2.5 and PM10
Year-wise Average PM2.5 Trend.
Average PM2.5 Contribution by Location.
55.32
Location
Visualizations
Build visual
Data
Search
wholefile
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1
Page 2
Page 3
Page 4
Page 5
Page 1 of 5
50%
Update available (click to download)

10. TABLEAU DASHBOARD

Tableau was used to visualize the air-quality dataset containing pollution measurements from different monitoring locations across India between 2020 and 2024. The dashboard presents four visualizations to compare pollution levels, analyze yearly trends, and understand variations in air quality. These visualizations make the data easier to interpret and support the identification of pollution patterns.

Year-wise Average PM2.5 Pollution Trends by Location

Problem Statement: How did average PM2.5 pollution levels change across different locations from 2020 to 2024?

Description: A line chart was created using the year from the Timestamp column on the X-axis and average PM2.5 concentration on the Y-axis. Different coloured lines represent individual monitoring locations.

Purpose: To identify yearly changes and compare pollution trends across locations.

Average PM2.5 by Location

Problem Statement: Which monitoring locations have higher average PM2.5 levels?

Description: A horizontal bar chart compares average PM2.5 concentrations across monitoring locations. The length of each bar represents the average pollution level.

Purpose: To identify locations with relatively high and low PM2.5 concentrations.

Average CO by Location

Problem Statement: How do carbon monoxide (CO) levels vary across monitoring locations?

Description: A vertical bar chart displays the average CO concentration for each location. The bar heights indicate differences in average CO levels.

Purpose: To compare carbon monoxide levels and identify locations with relatively higher concentrations.

Figure 3.3: Average CO by Location.

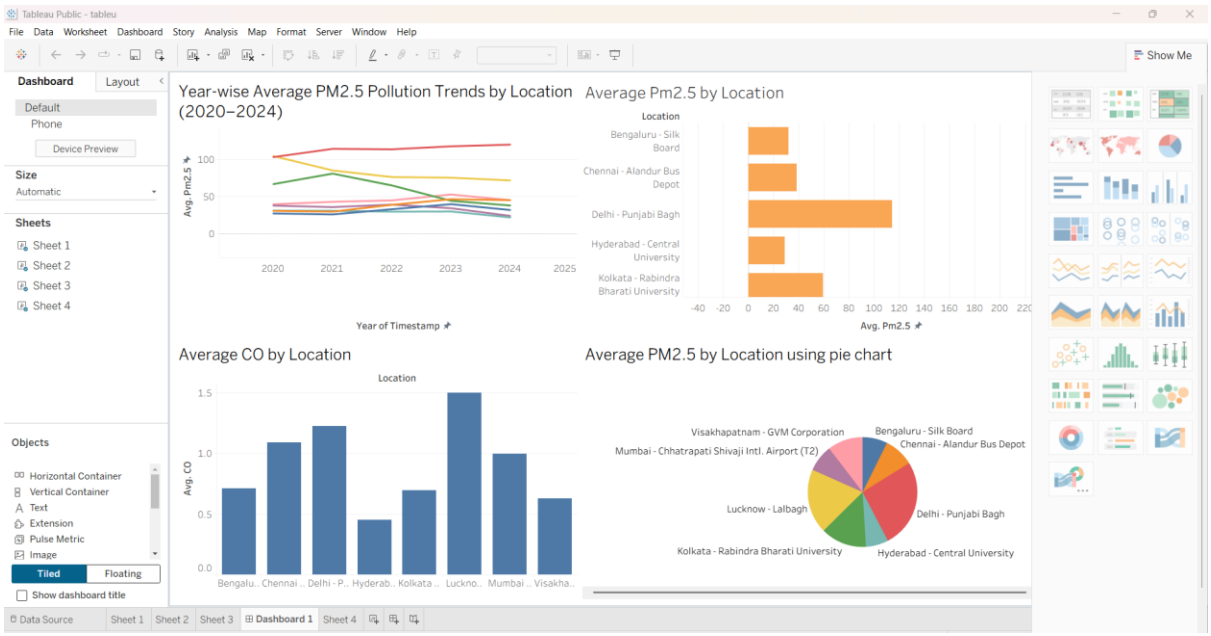
Average PM2.5 by Location Using a Pie Chart

Problem Statement: How do average PM2.5 levels compare among different monitoring locations?

Description: A pie chart displays each location's average PM2.5 concentration using a different coloured segment.

Purpose: To provide a visual comparison of average PM2.5 levels across the selected locations.

OUTPUT:



11. COMPARISON OF TOOLS

S.no	Basis of Comparison	Power BI	Tableau
1.	Main Purpose	To analyze pollution levels and visualize relationships between pollutants.	To compare pollution levels and explore variations across locations and years.
2.	Number of Visualizations	4	4
3.	Visualization Types	Clustered bar, line, scatter, and donut charts.	Line, horizontal bar, vertical bar, and pie charts.
4.	Main Pollutant Analyzed	PM2.5 and PM10	PM2.5 and CO
5.	Time-Based Analysis	Analyzes yearly PM2.5 trends from 2020 to 2024.	Compares yearly PM2.5 trends across multiple locations from 2020 to 2024.
6.	Location-Based Analysis	Compares average PM2.5 levels across locations.	Compares average PM2.5 and CO levels across locations.
7.	Key Benefit	Helps identify pollution patterns and relationships between pollutants.	Helps compare pollution variations across locations and over time.
8.	Overall Outcome	Supports the identification of pollution levels and pollutant relationships.	Supports the identification of location-wise pollution differences and yearly trends.

12. FINDINGS

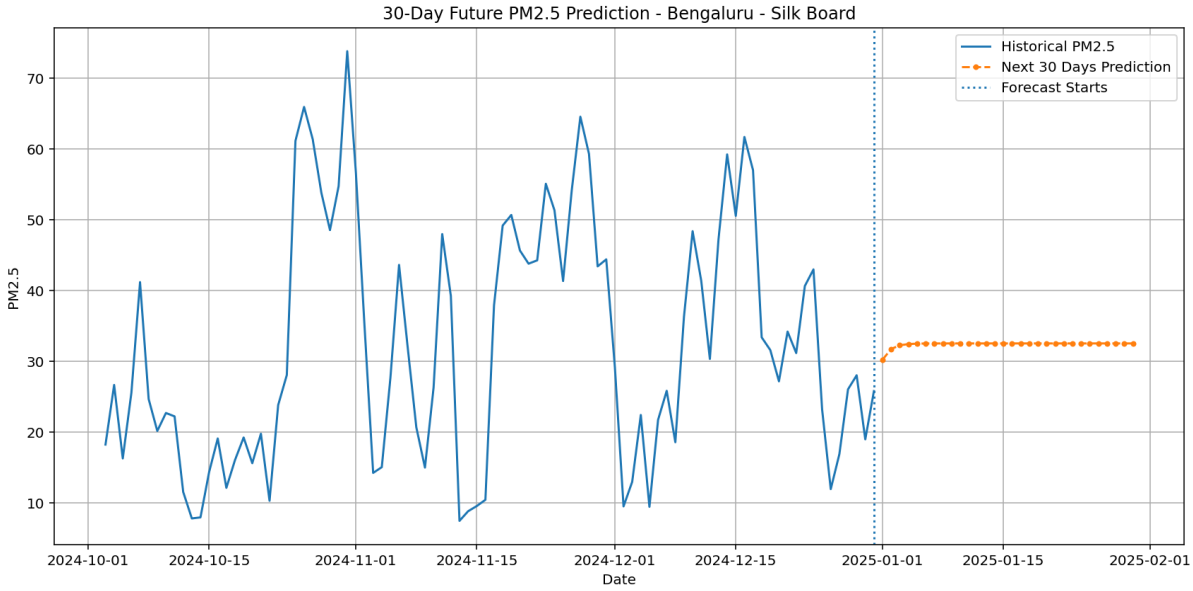
The ARIMA (AutoRegressive Integrated Moving Average) model was used to predict PM2.5 pollution levels for the next 30 days at different monitoring locations.

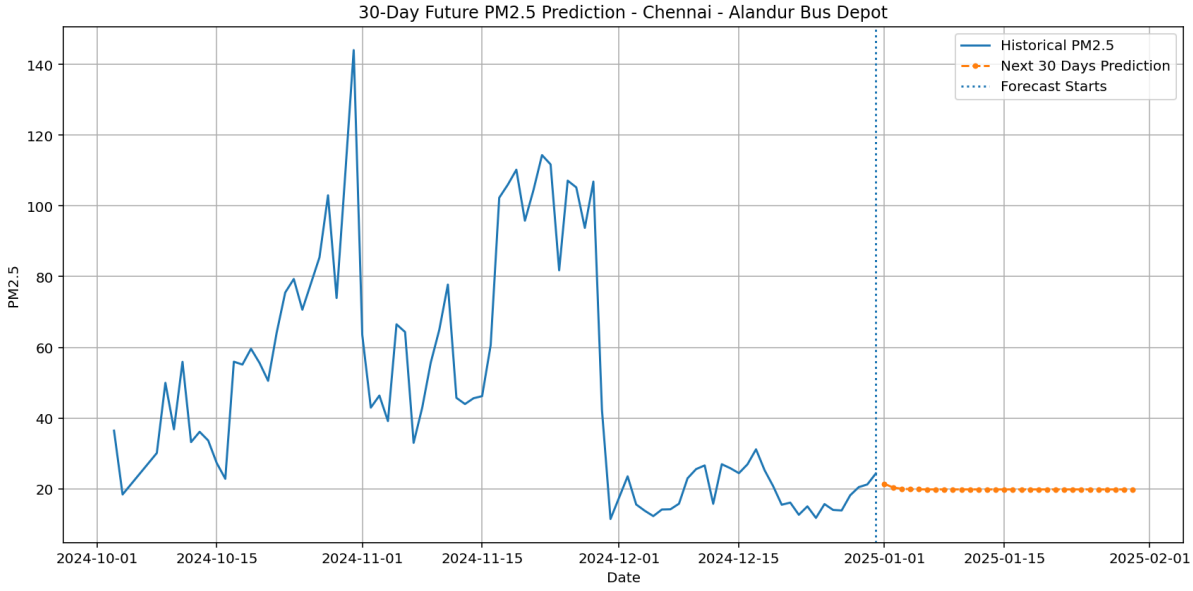
The historical data was divided into 80% training data and 20% testing data. The model was trained using the training data and evaluated using MAE, RMSE, and MAPE to measure prediction errors.

After evaluation, a new ARIMA model was trained using the complete historical data to predict PM2.5 levels for the next 30 days. The predicted values were saved in CSV files along with their corresponding dates and locations.

Graphs were generated to compare recent historical PM2.5 measurements with the predicted values. The evaluation results and future predictions were also combined into separate CSV files for further analysis.

Overall, this analysis helps examine predicted PM2.5 levels over the next 30 days and compare forecasts across different monitoring locations.





13. CONCLUSION

The project Air Quality Analysis, Pollution Prediction and Trend Forecasting was successfully carried out using air-quality data collected from different monitoring locations across India from 2020 to 2024. The project analyzed major air pollutants, including PM2.5, PM10, NO2, NH3, SO2, CO, and O3, to understand pollution levels and variations across locations and years.

Python was used for data processing, analysis, visualization, and forecasting. Power BI and Tableau were used to create charts and dashboards to compare pollution levels, identify yearly trends, and examine relationships between pollutants. The ARIMA model was also implemented to predict PM2.5 concentrations for the next 30 days. Model performance was evaluated using MAE, RMSE, and MAPE.

Overall, the project provides a clear understanding of historical air-quality patterns and estimated future PM2.5 levels. The visualizations and forecasting results can help identify areas with relatively higher pollution and support environmental monitoring and awareness. Although the predictions are estimates, the project demonstrates how data analytics and machine learning techniques can be used to study air pollution and support informed environmental decision-making.

14. REFERENCES

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